

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

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(MACRO-PRUDENTIAL AND MICRO-PRUDENTIAL REGULATIONS, BASEL I, II, III,
FED, ECB)

Source:

- *Introduction to banking*, Barabara Casu, Ch. 7
- *Risk management and financial institutions*, Hull

The *financial sector is one of the most heavily regulated sectors* in the economy and *banking is the most heavily regulated industry*. This should not be a surprise, since banking industry largely relies on public confidence, so banks are more prone to troubles than other firms, which can lead to bank runs. The banking system is *vulnerable to systematic risk*, and the risk that problems in one bank will spread through the whole sector (*contagion effect*).

Financial regulations, often enacted in response to a crisis, have the goal to ensure *financial stability*, *protect customers*, and *maintain confidence in the banking system*.

Systematic stability is one of the main reasons for regulation, as the social costs of bank failure are greater than the private costs. The second concern is to maintain the *safety and soundness of individual financial institutions*. The final reason relates to *consumer/depositor protection*.

Regulators want to ensure that banks keep enough capital for the risks they take. The fundamental question for banks is therefore the measurement of *capital adequacy*: do they have enough capital to cover future losses? The problem is with systematic risk, i.e. the risk that the failure by a large bank will lead to failure by other large banks and finally the financial system.

Macro- and micro-prudential regulation

Macro- and micro-prudential regulation is typically the responsibility of central banks (ECB, FED), financial supervisory authorities, and international organizations.

Systematic (macro-prudential) regulation

Macro-prudential regulation aims to *strengthen the resilience of the financial system* (mitigate systematic risk) and to *limit the built-up of systemic financial risks*. It is a *top-down approach*,

BANKING REGULATION AND SUPERVISION

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meaning that it considers the common exposure and correlation among institutions; it focuses more on the financial system as a whole, rather than on individual institutions. It also aims to ensure that *systematically important financial institutions* (SIFIs) are more heavily regulated and supervised.

Examples for tools of Macro-Prudential Regulation:

- *Counter-Cyclical Capital Buffer (CCyB)*: requires banks to hold additional capital during economic booms to absorb losses during downturns
- Cap on Loan-to-Value or Debt-to-Income ratios
- *Liquidity Coverage Ratio (LCR)*: ensures banks hold enough liquid assets to meet short-term obligations.

It is also concerned with *minimizing the risk of bank runs*. This is implemented through the *financial safety net*:

- Deposit insurance: aims to protect depositors
 - EU: all deposits up to €100 000 are protected through national DGS (Deposit Guarantee Schemes); European Deposit Insurance Scheme (EDIS) not yet implemented
 - US: deposits are automatically insured to at least \$250,000 at each FDIC-insured bank.
- Lender of last resorts (LOLR): bailing out SIFIs
 - Central banks act as a lender of last resort (LOLR) by providing emergency funds to banks facing collapse (solvent – the company can meet its obligations on the long run - but illiquid – cannot turn assets into cash quickly to meet short-term obligations - banks with good collateral). This prevents financial panic but can encourage risky behavior by banks who might expect bailouts. The 2008 financial crisis blurred the lines of LOLR, with central banks acting more like market makers of last resort (MMOLR) or ‘purchaser of last resort’. This expanded role is debated, with some concerned about inflation, transparency, and bailing out insolvent banks.

The ECB is responsible for assessing macroprudential measures adopted by national authorities in the countries participating in European Banking Supervision under the *Single Supervisory Mechanism (SSM) Regulation*.

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

Micro-prudential regulation

The objective of micro-prudential regulation is the *stability of individual financial institutions*. It is a *bottom-up approach*: correlations and common exposure across institutions are of secondary importance. *Micro-prudential supervision* checks how individual financial firms are complying with financial regulation and involves the collection and analysis of information about the risks that the firms take.

Examples for tools of Micro-Prudential Regulation:

- Capital Requirements: to cover potential losses
- Stress Testing
- Supervision: on-site inspection and off-site monitoring

Conduct of business regulation

Focuses on *how banks and other financial institutions conduct their business*. This kind of regulation relates to information disclosure, fair business practices, competence, honesty and integrity of financial institutions and their employees.

Micro- and macro-prudential policies usually complement each other in pursuit of their respective goals. However, conflict might occur because of overlapping policy mandates and the way in which policies are applied. Both micro- prudential and macro-prudential authorities might use the same policy instruments, but they have different objectives.

Basel regulations

Prior to Basel I, regulations set a minimum level for the ratio of capital to total assets, but these varied from country to country. Another problem was that over time banks began entering into more complicated transactions. As a response, a global regulatory framework was developed by the *Basel Committee on Banking Supervision (BCBS)*.

1) Basel I (1988)

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

It set international risk-based standards for *capital adequacy*. aimed to ensure that banks had sufficient capital to cover potential *credit losses*.

● *Capital Adequacy Ratio*

- ratio of a bank's capital to its *risk-weighted assets (RWAs)* - banks were required to assign a risk weight to each asset class, reflecting the asset's credit risk
- capital consisted of *Tier 1* (common stock, RE) and *Tier 2* (subordinated debt) capital

● *minimum capital requirement*

- at least 8% of risk-weighted assets (RWA), at least half of it coming from Tier 1 capital

Critics:

- too simple: ad hoc weights – not reflecting the risk associated with assets; assets within a category had same weight even if different in risk
- Deals only with credit risk
- The question "what is capital" is more important than "how much should capital be?,,

2) 1996 Amendment

- Introduced *capital for market risk* as well:
 - Standardized approach (simpler banks): regulators specified how to compute market risk; capital assigned separately to each type of market exposure (debt securities, equity securities, FX risk, commodity risk and option)
 - Internal Model Approach (more sophisticated banks): calculating VaR (*10-day 99% confidence level*) and converting it to capital requirement
- Required *VaR back-testing*: calculating the 1-day 99% VaR over the previous 250 days and counting exceptions to see whether VaR is over- or under-estimated.
- The total capital required became the sum of the 8% capital from credit risk and the capital from market risk.

3) Basel II (2004, implemented in 2007)

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

It is a more risk-sensitive approach to credit risk management and capital adequacy requirements.

In the US it applied only to large banks, in the EU to all banks. It has three pillars:

- **New minimum capital requirements for credit and operational risk**

- Credit risk: Standardized Approach vs Internal Ratings-Based (IRB) Approach
- Separate capital charge for operational risk: BIS, STA, AMA

- **Supervisory review**

- Supervisors ensure that capital levels are maintained
- More emphasis on early intervention when problems arise

- **Market discipline**

- Focuses on transparency by increasing the disclosure of risk assessment procedures and capital adequacy

- The total capital required became the sum of the 8% capital from credit risk, operational risk and market risk.

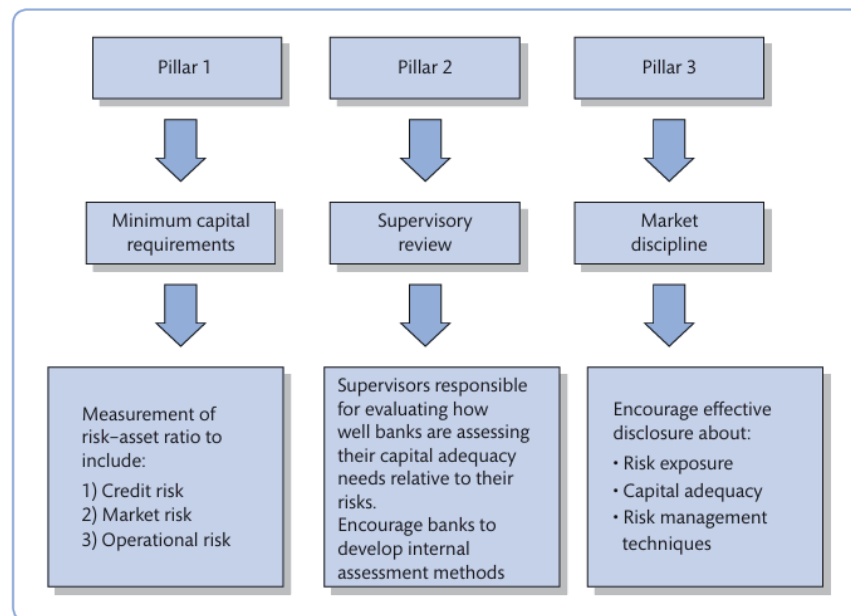


Figure 7.5 Basel II: the three pillars

Critics:

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

- *Pro-cyclical*: the regulation may worsen downturns (as credit risk increases, capital requirements rose), while during booms capital requirements were low, encouraging excessive risk-taking

Credit risk measurement methods: Standard Approach, Foundation IRB Approach, Advanced IRB Approach.

● Standard method (STA)

- Used by smaller, simple banks
- Risk weights from Basel I were allowed to be based on external credit ratings

● Internal Rating Based (IRB) method

- **Foundation IRB**: Banks estimate certain inputs (e.g., probability of default) while others are provided by regulators.
- **Advanced IRB**: Banks estimate all key inputs themselves.
- Capital requirement is the *1-year 99.9% confidence level VaR – EL*:

$$\sum_i \text{EAD}_i \times \text{LGD}_i \times (\text{WCDR}_i - \text{PD}_i)$$

- The amount of capital required is given by the *capital function*:

$$K = (\text{LGD} \times N(x) - \text{LGD} \times \text{PD}) \times (1 + (M - 2.5) b) / (1 - 1.5 b)$$

Operational risk is the *risk of loss resulting from inadequate or failed internal processes, people and systems, or from external events*. Operational risk measurement methods:

- **Basic indicator approach (BIA)**: 15% of average annual gross income over the previous 3 years
- **Standardized approach (STA)**: average annual gross income over the last 3 years of the eight business lines multiplied by beta factors and summed up to get the total capital
- **Advanced measurement approach (AMA)**: bank's own internal models, *1-year 99.9% CI VaR*

Which of them is used is based on the sophistication of the bank.

4) Basel III (2010)

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

- stricter capital requirements and liquidity standards in response to the GFC of 2008
- increasing the quality and quantity of capital banks are required to hold to withstand periods of stress

It has six parts:

- **Higher minimum Tier 1 capital**

- Capital consists of **Tier 1 equity capital** (common equity), **additional Tier 1 capital** (cumulative preferred stocks) and **Tier 2 capital** (subordinated debt)
- Tier 1 capital ratio to RWAs increased from 4% to 6 %
- In total still at least 8% of RWA

- **Capital Conservation Buffer**

- Extra **2.5% of RWA** (common equity) required in normal times to absorb losses in periods of financial and economic stress; total thus becomes *10.5% of RWA*

- **Countercyclical Buffer**

- Extra equity capital, to allow for cyclicalities of bank earnings, between **0-2.5% of RWA** (common equity)
- Left to the discretion of national regulators

- **Minimum Leverage Ratio** is one of the most important innovations from a credit risk perspective – ratio of a capital measure to an exposure measure

- Ratio of **Tier 1 capital to Total Exposure** must be **at least 3%**
- Exposure includes all on-balance sheet items, derivatives, securities financing exposures, and some off-balance sheet items

- **Liquidity Ratios**

- *Liquidity risk* arises because there is a tendency for banks to finance long-term needs with short-term funding
- **Liquidity Coverage Ratio**: ability to survive a 30-day period of liquidity disruptions

$$\text{Liquidity Coverage Ratio} = \frac{\text{High Quality Liquid Assets}}{\text{Net Cash Outflows for 30 day period}} \geq 100\%$$

- **Net Stable Funding Ratio**: liquidity management over a period of one year

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

$$\text{Net Stable Funding Ratio} = \frac{\text{Amount of Stable Funding}}{\text{Required Amount of Stable Funding}} \geq 100\%$$

- Ratios should be greater than 100%

- **Counterparty credit risk**

- **Credit Value Adjustment** (CVA) calculated for each derivative counterparty (exp loss bc of counterparty default); reported profits are reduced by the total of the CVAs

ECB

The European Central Bank (ECB) is the central bank responsible for managing the euro, the common currency used by 20 of the 27 European Union (EU) member states, collectively known as the Eurozone. The ECB's primary objective is to *maintain price stability within the Eurozone*, which it defines as keeping inflation rates below, but close to, 2% over the medium term. The primary monetary policy instrument is the setting of ECB policy rates. It also aims to ensure the *smooth operation of payment systems*.

Location: headquartered in Frankfurt

Legal Basis: The ECB operates under the Treaty on the Functioning of the European Union (TFEU) and the Statute of the European System of Central Banks (ESCB) and the ECB. It is an independent institution, meaning *it makes decisions free from political influence*.

Structure and Governance

Executive Board:

Consists of the President, Vice-President, and four other members. The board is responsible for *implementing monetary policy and managing the day-to-day operations* of the ECB.

Governing Council:

The main *decision-making body*, comprising the members of the Executive Board and the governors of the national central banks of the Eurozone countries. It sets the monetary policy direction and makes key decisions.

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

General Council:

Includes the President and Vice-President of the ECB and the governors of all EU member states' national central banks, playing a *consultative role*.

Supervisory Board:

Oversees banking supervision within the framework of the SSM, ensuring banks adhere to regulatory standards.

In March 2013 the European Parliament and European Council agreed on regulation regarding the interaction of the EBA and the European Central Bank's (ECB) prudential supervision powers within the new SSM. The regulation came into force in November 2013. The transfer of supervisory tasks to the ECB is based on Article 127(6) of the Treaty on the Functioning of the European Union (also known as the Lisbon Treaty). More specifically, this regulation details:

- new competencies of the European Central Bank as supervisor;
- co-operation between the European Central Bank and the European Banking Authority.

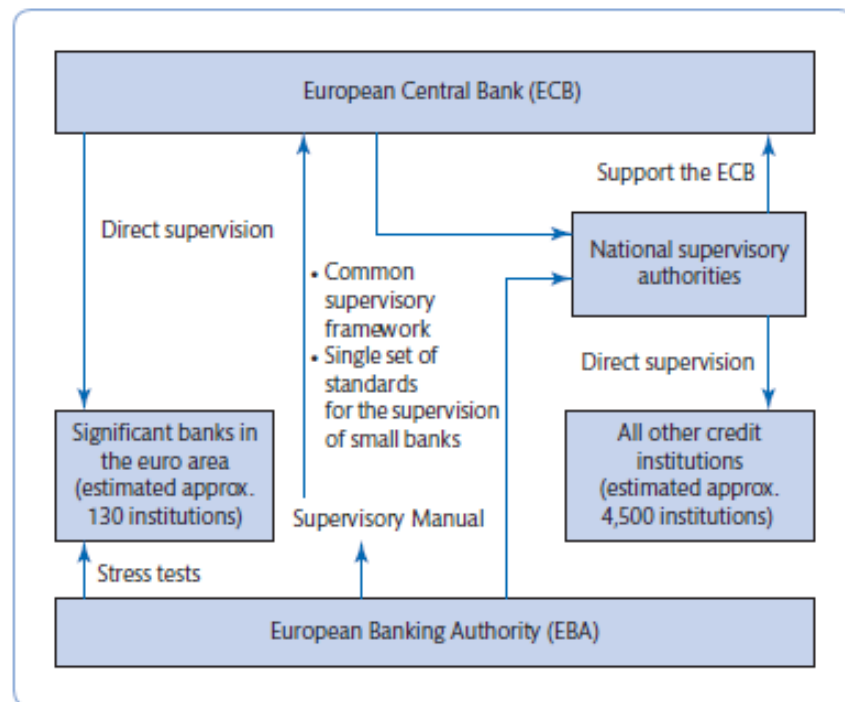


Figure 14.7 The Single Supervisory Mechanism

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

FED

The Federal Reserve, commonly referred to as the Fed, is the central bank of the United States. It was established in 1913 with the primary objective of providing the country with a safe, flexible, and stable monetary and financial system. It has a dual mandate to achieve *maximum employment* and *price stability*, while also maintaining *financial stability*.

Structure and Governance

Board of Governors:

Comprises seven members appointed by the President of the United States and confirmed by the Senate. The Board *oversees the Federal Reserve System*, *guides monetary policy*, and *has regulatory and supervisory responsibilities*.

Federal Open Market Committee (FOMC):

Consists of the Board of Governors and five of the twelve Federal Reserve Bank presidents. The FOMC *sets monetary policy*, including the target for the federal funds rate and the direction of open market operations.

Federal Reserve Banks:

There are *twelve regional Federal Reserve Banks located in major cities* across the U.S. These banks *implement Fed policies at the regional level*, provide financial services, and gather economic data.

Typically, the Federal Reserve is responsible for *regulating the largest and most complex US banks* as it has responsibility for bank and diversified financial holding companies. It also regulates the operations of foreign banks, many of which are engaged in relatively complicated investment and corporate banking activities.

The 2010 Dodd–Frank Act reformed the Federal Reserve. The Act made the Fed the principal *regulator for systemically important financial institutions*. The emergency lending powers of the Federal Reserve give it authority to lend to non-depository institutions (investment banks, insurers) in ‘unusual and exigent circumstances’. Dodd–Frank now makes this lending conditional on approval from the Secretary of the Treasury. The Federal Reserve will have to disclose all its Section 13(3) lending details and will be subject to a one-time Government Accountability Office

BANKING REGULATION AND SUPERVISION

(macro-prudential and micro-prudential regulations, Basel I, II, III, FED, ECB)

(GAO) audit of emergency lending during the crises times. A new post of the Federal Reserve Vice Chair for supervision will be created and the GAO will also examine governance issues and how directors are appointed.

The FED conducts annual Comprehensive Capital Analysis and Review (CCAR) and Dodd-Frank Act Stress Tests (DFAST) to assess large banks' ability to handle economic shocks. These tests go beyond Basel III, ensuring that banks maintain adequate capital buffers under stressed scenarios.